Finding the Most Profitable Investment in APAC Using Python

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Python APAC 2024

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Introduction

Definitions

What does profitable mean?

- Yield
- Risk-to-reward: Sharpe ratio

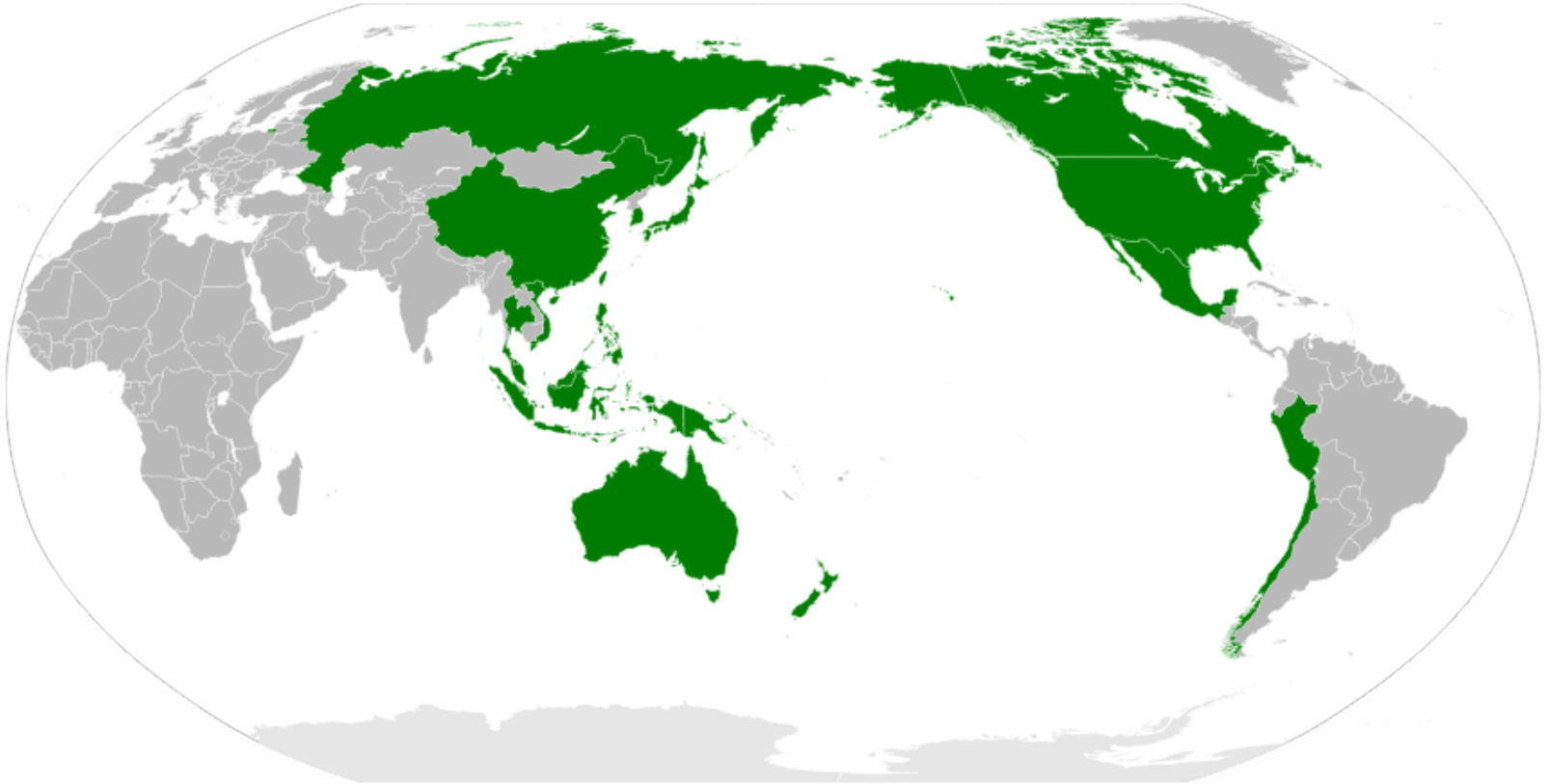
Definitions

Assets:

- Stocks
- ETFs
- Cryptocurrencies: BTC & ETH

Definitions

APAC -> Region of countries adjoining the Pacific Ocean



Definitions

APEC members:

- Australia
- Brunei Darussalam
- Canada
- Indonesia
- Japan
- South Korea
- Malaysia
- New Zealand
- Philippines
- Singapore
- Thailand
- United States
- Taiwan
- Hong Kong
- China
- Mexico
- Papua New Guinea
- Chile
- Peru
- Russia
- Vietnam
- India (bonus)

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Objective

Find the 10 best performing, publicly available assets in APAC based on yield and risk-to-reward ratio

Methodology

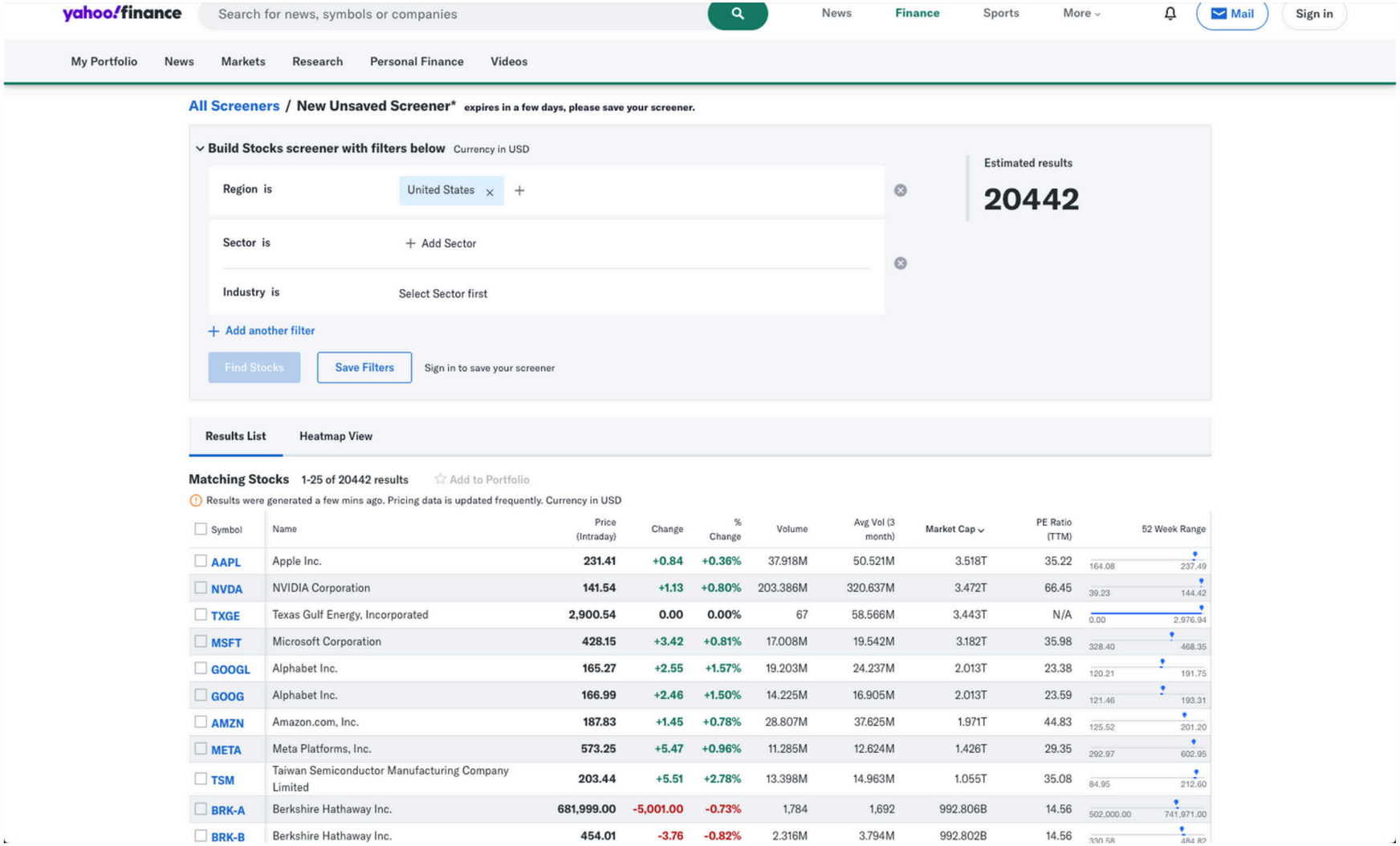
Setup

```
import numpy as np
import pandas as pd
import requests
import yfinance as yf

from tqdm.notebook import tqdm
```

Scrape Yahoo Finance

for asset tickers list



Symbol	Name	Price (Intraday)	Change	% Change	Volume	Avg Vol (3 month)	Market Cap	PE Ratio (TTM)	52 Week Range
AAPL	Apple Inc.	231.41	+0.84	+0.36%	37.918M	50.521M	3.518T	35.22	164.08 237.49
NVDA	NVIDIA Corporation	141.54	+1.13	+0.80%	203.386M	320.637M	3.472T	66.45	39.23 144.42
TXGE	Texas Gulf Energy, Incorporated	2,900.54	0.00	0.00%	67	58.566M	3.443T	N/A	0.00 2,976.94
MSFT	Microsoft Corporation	428.15	+3.42	+0.81%	17.008M	19.542M	3.182T	35.98	328.40 468.35
GOOGL	Alphabet Inc.	165.27	+2.55	+1.57%	19.203M	24.237M	2.013T	23.38	120.21 191.75
GOOG	Alphabet Inc.	166.99	+2.46	+1.50%	14.225M	16.905M	2.013T	23.59	121.46 193.31
AMZN	Amazon.com, Inc.	187.83	+1.45	+0.78%	28.807M	37.625M	1.971T	44.83	125.52 201.20
META	Meta Platforms, Inc.	573.25	+5.47	+0.96%	11.285M	12.624M	1.426T	29.35	292.97 602.95
TSM	Taiwan Semiconductor Manufacturing Company Limited	203.44	+5.51	+2.78%	13.398M	14.963M	1.055T	35.08	84.95 212.60
BRK-A	Berkshire Hathaway Inc.	681,999.00	-5,001.00	-0.73%	1,784	1,692	992.806B	14.56	502,000.00 741,971.00
BRK-B	Berkshire Hathaway Inc.	454.01	-3.76	-0.82%	2.316M	3.794M	992.802B	14.56	330.58 484.82

Scrape Yahoo Finance

```
# Yahoo Finance exchange codes for each country
region_list = [
    "au", # Australia
    "ca", # Canada
    "id", # Indonesia
    "jp", # Japan
    "kr", # South Korea
    "my", # Malaysia
    "nz", # New Zealand
    "sg", # Singapore
    "th", # Thailand
    "us", # United States
    "tw", # Taiwan
    "hk", # Hong Kong
    "cn", # China
    "mx", # Mexico
    "cl", # Chile
    "in", # India
]
```

```
# Yahoo Finance equity sectors
sector_list = [
    "Basic Materials",
    "Financial Services",
    "Consumer Defensive",
    "Utilities",
    "Energy",
    "Technology",
    "Consumer Cyclical",
    "Real Estate",
    "Healthcare",
    "Communication Services",
    "Industrials"
]
```

Scrape Yahoo Finance

```
base_url = "https://query1.finance.yahoo.com/v1/finance/screener"

headers = {
    "cookie": "gpp=DBAA; gpp_sid=-1;",
    "user-agent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36",
}

params = {
    "crumb": "7ms989myAtW",
    "lang": "en-US",
    "region": "US",
    "formatted": "true",
    "corsDomain": "finance.yahoo.com"
}

size = 250 # Max size per request

payload_template = {
    "size": size,
    "sortField": "ticker",
    "sortType": "asc",
    "topOperator": "AND",
    "userId": "",
    "userIdType": "guid"
}


tickers = set()
```

Scrape Yahoo Finance

```
def fetch_tickers(quote_type, operands):
    global size
    global tickers

    offset = 0
    total_tickers = 0

    while True:
        # Update the offset in the payload for each request
        payload = {
            **payload_template,
            "quoteType": quote_type,
            "query": {
                "operator": "AND",
                "operands": operands
            },
            "offset": offset
        }

        # Make the POST request
        with requests.session() as session:
            # Set cookies
            session.cookies['A1'] = headers['cookie']
            response = session.post(
                base_url,
                params=params,
                json=payload,
                headers=headers
            )
```

Scrape Yahoo Finance

```
# Check if the request was successful
if response.status_code == 200:
    data = response.json()
    # Extract symbols and add them to the tickers set
    quotes = data.get("finance", {}).get("result", [{}])[0].get("quotes", [])
    for quote in quotes:
        tickers.add(quote["symbol"])

    total_tickers = data.get("finance", {}).get("result", [{}])[0].get("total", 0)
    print(f"Retrieved {len(quotes)} tickers, offset: {payload['offset']}, tickers_length: {len(tickers)}, total: {total_tickers}")

    # Increase offset for the next request
    offset += size

    # Break if all tickers have been retrieved
    if offset >= total_tickers:
        break
else:
    print(f"Error: {response.status_code} - {response.text}")
    break  # Exit on error
```

Scrape Yahoo Finance

```python
def fetch_equity_tickers():
    # Fetch tickers for each combination of region and sector
    for region in region_list:
        for sector in sector_list:
            operands = [
                {"operator": "or", "operands": [{"operator": "EQ", "operands": ["region", region]}]},
                {"operator": "or", "operands": [{"operator": "EQ", "operands": ["sector", sector]}]}
            ]

            fetch_tickers("EQUITY", operands)
```

```python
def fetch_etf_tickers():
    operands = [
        {"operator": "or", "operands": [{"operator": "EQ", "operands": ["region", region]} for region in region_list]}
    ]

    fetch_tickers("ETF", operands)
```

```python
# Fetch equity tickers
fetch_equity_tickers()
# Fetch ETF tickers
fetch_etf_tickers()
```

Add BTC and ETH Tickers

```
cryptocurrency_tickers = {"BTC-USD", "ETH-USD"}
tickers.update(cryptocurrency_tickers)
```

```
print(len(tickers))
print(tickers)
```

```
58429
```

```
'BTCY-B.TO', 'CIMEF', 'GOOD', 'MSBIP', '2285.HK', '002560.SZ', '1662.HK', 'SPOT', '603083.SS', '5242.T', 'VCIG',
'BCDMF', 'PLLR.CN', 'SLMXF', '688286.SS', 'BARUF', 'ISDR', 'THP.V', 'KANSHST.BO', 'TNEN', 'HDIH', 'LAFFANSQ.BO',
, 'RFL', 'FW.V', '6277.TW', 'HUTE.TO', 'FSRNQ', 'CNGA', '459370.KS', 'ARKK', '601600.SS', 'IEGCF', '8143.HK', '
NSBF', 'TIPX', '2246.T', 'HYMCL', '8627.HK', '6050.T', 'EMAX', 'ANRG.TO', 'INUV', 'CENTRUM.NS', 'PSSR', '377450.
.NS', 'GEBHF', 'HSTXF', '139660.KS', 'OCFCP', 'WM.TO', '4316.KL', 'FFF.CN', 'WLD.AX', '1712.HK', 'ATCD', 'VATE',
', 'GAME.XA', '017960.KS', '6899.TWO', '138610.KQ', 'ALIF', 'TQR-R.BK', 'NZRFF', '9411.HK', '5130.KL', 'IMMVF',
'SORHF', '688015.SS', 'AAA.AX', 'JCTC', 'GAS.AX', 'FJTSF', 'FWDG', 'SINI.JK', 'VI.TO', '3814.T', 'LALIN-R.BK',
NFO.NS', '2434.TW', 'DAN.V', 'PNRG', '4510.TWO', 'JETFREIGHT.BO', '00785B.TWO', 'CBRL', 'ATUL.BO', '252420.KS',
90.KS', '6434.TWO', '432320.KS', 'BEKE', 'CETY', '5803.T', 'SGU.V', 'BRLAF', '1220.HK', 'BWF.AX', 'XJAN', 'NFLX
', 'DYAI', 'COHG', 'XL.V', 'SCC-R.BK', 'LASR', 'TAYS.JK', 'CLSEL.BO', '301050.SZ', 'GMMPFAUDLR.NS', '3861.T', '
TGS', '2119.HK', 'GRQ.SI', 'ASKAUTOLTD.BO', 'SUPRAJIT.NS', '570109.KS', 'PAMC', '000811.SZ', 'SKY.AX', '8208.T',
CH.BO', 'IXIGO.BO', 'PHYD', 'NXTYQ', '6222.TWO', '2333.HK', '300675.SZ', 'PCHM', '378850.KS', 'CBDG', 'SAB.BO',
.T', 'AMNTF', '0034.HK', 'IRMTF', '300722.SZ', 'AUGB-F.TO', 'DBRG-PJ', 'BKNIY', 'LEA', 'BBSA', 'SSBI', 'RAYJ',
-R.BK', 'SFIX', 'TCSB.TO', 'AQUA.BK', 'ATDRF', '3388.TWO', '6823.HK', '688049.SS', '016380.KS', '00753L.TW', 'SI
 'ASPINWALL.NS', 'UDIV-U.TO', 'MNUFF', 'EGHT', 'STCK.TO', 'PTL.NS', 'AETH.CN', 'DECK', '603659.SS', 'IBATF', '6
 '2539.TW', 'SPZ.V', 'GTHR', 'TEXELIN.BO', 'VANTABIO.BO', '6083.HK', 'ISEC.BO', 'QTRH.TO', '002683.SZ', '6016.T
'JSWINFRA.NS', '301087.SZ', '600580.SS', 'NCNC', 'SPXL.MX', 'XOS', 'MHKI.JK', '317330.KQ', 'TOP.BK', 'FFIC', 'PS
PLY.BO', 'EPGG', 'RSG', '0526.HK', 'KG.V', '300122.SZ', 'KRNTY', '001560.KS', '1626.TW', '301310.SZ', 'MIDHANI.N
02745.SZ', '8066.TWO', 'FELTF', '0243.HK', 'UNP', 'MA.NE', 'HXS-U.TO', 'IEHC', 'COD.AX', '7616.T', 'CSPUF', 'IWF
7562.T', 'DBI.AX', '7611.T', '603612.SS', '3880.T', '302190.KS', 'BUTTERFLY.NS', '7609.T', 'AKS.BK', '301185.SZ
K', 'TRRPF', '3600.HK', 'AAUC.TO', 'GNE.NZ', 'RPX.V', '5445.T', '300629.SZ', 'TNPH', '00694B.TWO', 'KWEB.MX', '6
18.KL', 'CFG-PH', 'SVMHW', 'VSTTILLERS.BO', '0876.HK', 'CI.BK', 'BLUE', '0618.HK', 'SKIPPERPP.BO', '7989.T', '24
600586.SS', 'SFLEX-R.BK', 'CBGI', 'MTB.AX', 'VSB.TO', 'CV-R.BK', 'UNITDSPR.BO', 'CWNOF', 'SAPPE.BK', '002663.SZ
```

Calculate Return and Sharpe Ratio

```python
def calculate_asset_profitability(tickers, start_date, end_date, batch_size=100):
    """Batch download stock data with progress tracking"""
    tickers = list(tickers)
    asset_returns = {}
    asset_sharpe_ratios = {}

    for i in tqdm(range(0, len(tickers), batch_size)):
        batch = tickers[i:i + batch_size]
        batch_string = " ".join(batch)
        data = yf.download(
            batch_string,
            start=start_date,
            end=end_date,
            progress=False
        )

        returns, sharpe_ratios = calculate_return_and_sharpe(data)

        asset_returns.update(returns)
        asset_sharpe_ratios.update(sharpe_ratios)

    result_df = pd.DataFrame({
        'Returns': pd.Series(asset_returns),
        'Sharpe': pd.Series(asset_sharpe_ratios)
    })

    return result_df
```

Calculate Return and Sharpe Ratio

```python
def calculate_return_and_sharpe(data, risk_free_rate=0.0):
    returns = {}
    sharpe_ratios = {}

    # Calculate daily returns for each ticker in the batch
    daily_returns = data['Adj Close'].pct_change().dropna()

    # Calculate overall returns and Sharpe ratio for each ticker
    for ticker in daily_returns.columns:
        start_price = data['Adj Close'][ticker].iloc[0]
        end_price = data['Adj Close'][ticker].iloc[-1]
        overall_return = (end_price - start_price) / start_price
        returns[ticker] = overall_return

        # Calculate Sharpe ratio, assuming a risk-free rate of 0
        mean_daily_return = (daily_returns[ticker] - risk_free_rate).mean()
        std_dev = daily_returns[ticker].std() # standard deviation
        annual_trading_days = 252
        annualization_factor = np.sqrt(annual_trading_days)
        sharpe_ratio = mean_daily_return / std_dev * annualization_factor if std_dev != 0 else 0
        sharpe_ratios[ticker] = sharpe_ratio

    return returns, sharpe_ratios
```

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_p}$$

where:

R_p = return of portfolio

R_f = risk-free rate

σ_p = standard deviation of the portfolio's excess return

Calculate Return and Sharpe Ratio

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def calculate_asset_profitability(tickers, start_date, end_date, batch_size=100):
    """Batch download stock data with progress tracking"""
    tickers = list(tickers)
    asset_returns = {}
    asset_sharpe_ratios = {}

    for i in tqdm(range(0, len(tickers), batch_size)):
        batch = tickers[i:i + batch_size]
        batch_string = " ".join(batch)
        data = yf.download(
            batch_string,
            start=start_date,
            end=end_date,
            progress=False
        )

        returns, sharpe_ratios = calculate_return_and_sharpe(data)

        asset_returns.update(returns)
        asset_sharpe_ratios.update(sharpe_ratios)

    result_df = pd.DataFrame({
        'Returns': pd.Series(asset_returns),
        'Sharpe': pd.Series(asset_sharpe_ratios)
    })

    return result_df
```

Calculate Return and Sharpe Ratio

```
start_date = "2014-01-01"
end_date = "2024-09-30"

data = calculate_asset_profitability(
    tickers=tickers,
    start_date=start_date,
    end_date=end_date
)

print(data)
```

```
           Returns    Sharpe
0873.HK        NaN -0.558849
GTPS           NaN  0.306409
0104.KL        NaN  0.626777
6261.TWO       NaN  0.439885
NWPN           NaN  1.265332
0215.HK        NaN  0.011254
600619.SS      NaN  0.261552
290090.KQ      NaN -0.864296
8299.TWO       NaN  0.531027
WPNDF          NaN -0.343387
```

Final Step - Sort Our Data

```
highest_returns = data.sort_values(by='Returns', ascending=False).head(10)
highest_sharpe = data.dropna(subset=['Returns']).sort_values(by='Sharpe', ascending=False).head(10)
```

Findings

Top 10 Returns

	Ticker	Country	Sector	Returns	Sharpe
1	CUPID.BO	India	Consumer Defensive	5443.72	1.44
2	ZSARACOM.BO	India		2888.98	2
3	PRECWIRE.NS	India	Industrials	984.41	1.04
4	PRECWIRE.BO	India	Industrials	880.43	1.04
5	BEL.NS	India	Industrials	741.38	0.68
6	GRMOVER.BO	India	Consumer Defensive	722.85	1.19
7	BEL.BO	India	Industrials	719.26	0.67
8	JYOTIRES.BO	India		675.25	1.3
9	REFEX.NS	India	Energy	547.75	1.18
10	REFEX.BO	India	Energy	536.45	1.17

Top 10 Sharpe Ratio

	Ticker	Country	Sector	Sharpe	Returns
1	VTI.MX	Mexico	ETF	3.32	3.43
2	ZSARACOM.BO	India		2	2888.98
3	VISESHINFO.NS	India	Technology	1.79	0.17
4	CYBL	United States	Industrials	1.72	22
5	BAPACK.BO	India		1.67	174.85
6	GALXBRG.BO	India		1.62	379.4
7	CEENIK.BO	India		1.61	219.37
8	VERITAS.BO	India		1.56	40.8
9	KOTIC.BO	India		1.54	21.57
10	NATIONSTD.BO	India		1.54	258.25

Limitations

Considerations

- 6 APEC countries are excluded due to data unavailability
- Yahoo Finance may not have the complete ticker data
- Sharpe ratio calculation assumes risk-free rate is zero
- Historical data does not guarantee future performances

Conclusion

Summary

- We analyzed over 58000 assets; 50000 equity and 8000 ETFs
- 95 percent of the most profitable assets are equity, only one is an ETF
- 90 percent of the most profitable assets are Indian stocks
- ZSARACOM is a prominent asset, ranking second in both return and Sharpe Ratio metrics

References

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Thank You

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